

Initiating Search

November 4, 2025, 9:47 PM

• Search:

CAS Lexicon Search:

Gabriel synthesis - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (188)	 References	View Results
Exported: Retrieved Related Reaction Results (263)	 Reactions	View Results

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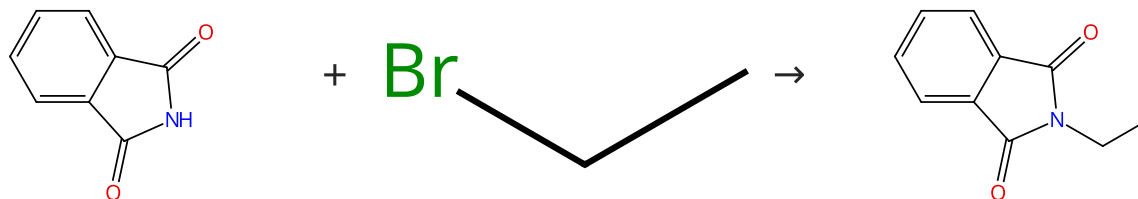
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (82)

 Suppliers (54)

 Suppliers (57)

31-041-CAS-9836009

Steps: 1 Yield: 100%

Effects of cryptands and activation of bases. V. Action of alkali hydrides on weak acids. II. Alkylation of anions obtained

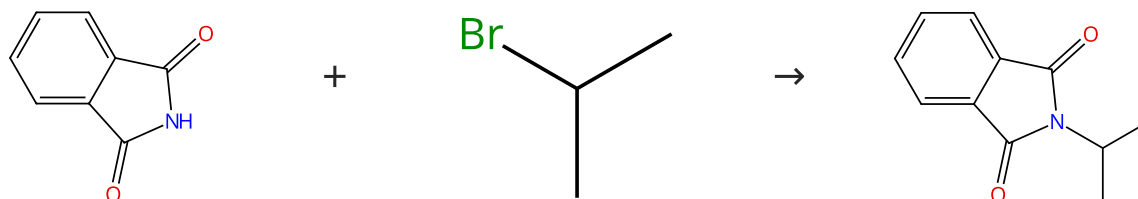
No Data Available

By: Pasquini, M. A.; et al

Tetrahedron (1980), 36(9), 1223-6.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (82)

 Suppliers (67)

 Suppliers (58)

31-041-CAS-4968509

Steps: 1 Yield: 100%

Effects of cryptands and activation of bases. V. Action of alkali hydrides on weak acids. II. Alkylation of anions obtained

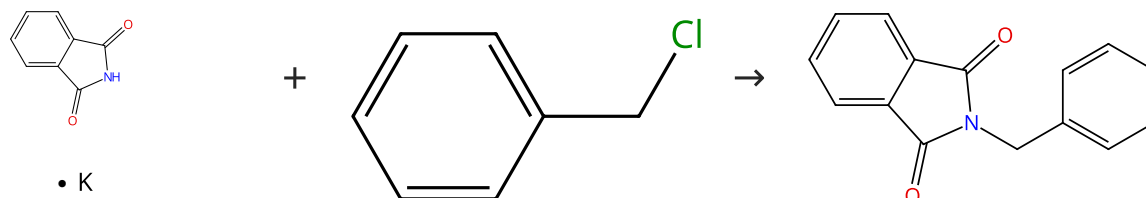
No Data Available

By: Pasquini, M. A.; et al

Tetrahedron (1980), 36(9), 1223-6.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (93)

 Suppliers (65)

 Suppliers (77)

31-041-CAS-16819426

Steps: 1 Yield: 100%

Phthalimide

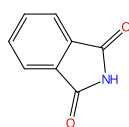
1.1 Catalysts: Amberlyst A 27
Solvents: Toluene; 90 °C

By: de Koning, Henk; et al

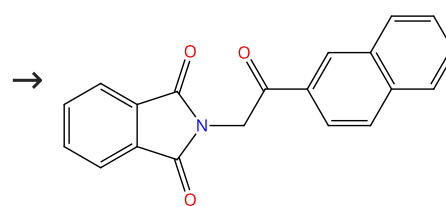
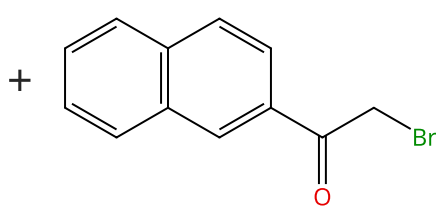
e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-3.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%



• K


 Suppliers (93)

 Suppliers (75)

 Suppliers (4)

31-041-CAS-12352332

Steps: 1 Yield: 99%

Organic solid-state reactions. Solid-state Gabriel reactions

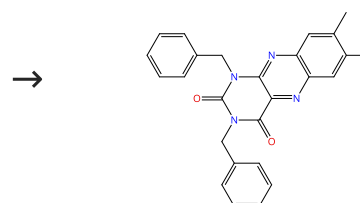
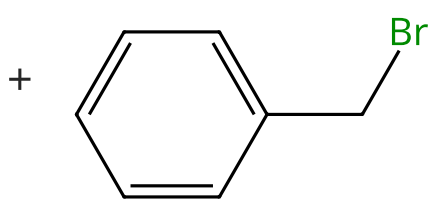
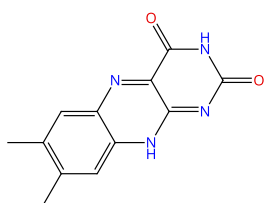
No Data Available

By: Tokuno, Kenji; et al

Yakugaku Zasshi (1991), 111(7), 359-64.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (79)

 Suppliers (67)

31-041-CAS-18435839

Steps: 1 Yield: 99%

Mechanochemical N-alkylation of imides

1.1 Reagents: Potassium carbonate; 1 h

By: Bris, Anamarija; et al

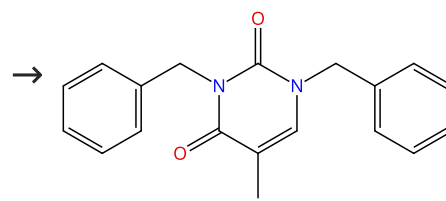
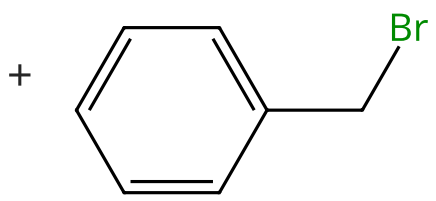
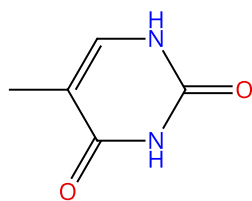
1.2 Solvents: Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (123)

 Suppliers (67)

 Suppliers (3)

31-041-CAS-18435834

Steps: 1 Yield: 99%

Mechanochemical N-alkylation of imides

1.1 Reagents: Potassium carbonate; 1 h

By: Bris, Anamarija; et al

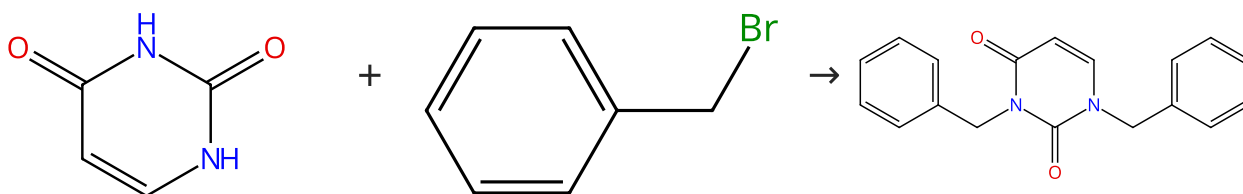
1.2 Solvents: Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (133)

Suppliers (67)

Suppliers (8)

31-041-CAS-18435835

Steps: 1 Yield: 98%

Mechanochemical N-alkylation of imides1.1 **Reagents:** Potassium carbonate; 1 h

By: Bris, Anamarija; et al

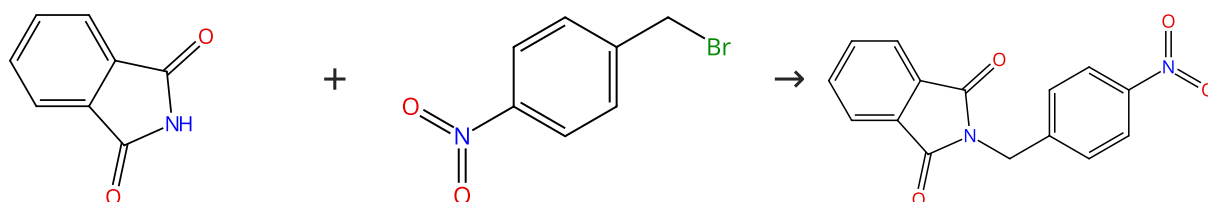
1.2 **Solvents:** Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (82)

Suppliers (68)

Suppliers (37)

31-041-CAS-18435829

Steps: 1 Yield: 98%

Mechanochemical N-alkylation of imides1.1 **Reagents:** Potassium carbonate; 1 h

By: Bris, Anamarija; et al

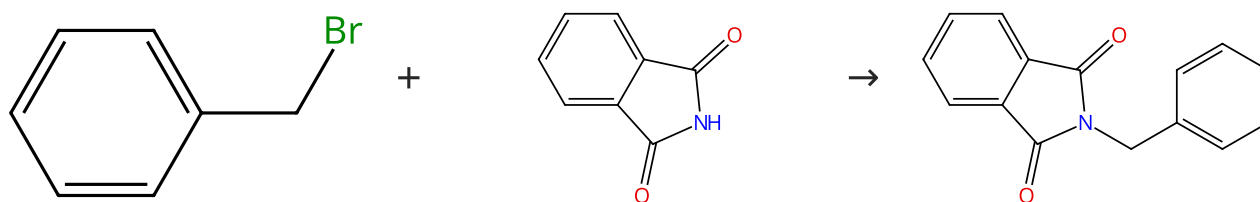
1.2 **Solvents:** Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 9 (2 Reactions)

Steps: 1 Yield: 97-98%



Suppliers (67)

Suppliers (82)

Suppliers (77)

31-041-CAS-8028052

Steps: 1 Yield: 98%

Effects of cryptands and activation of bases. V. Action of alkali hydrides on weak acids. II. Alkylation of anions obtained

No Data Available

By: Pasquini, M. A.; et al

Tetrahedron (1980), 36(9), 1223-6.

31-041-CAS-18435825

Steps: 1 Yield: 97%

Mechanochemical N-alkylation of imides1.1 **Reagents:** Potassium carbonate; 1 h

By: Bris, Anamarija; et al

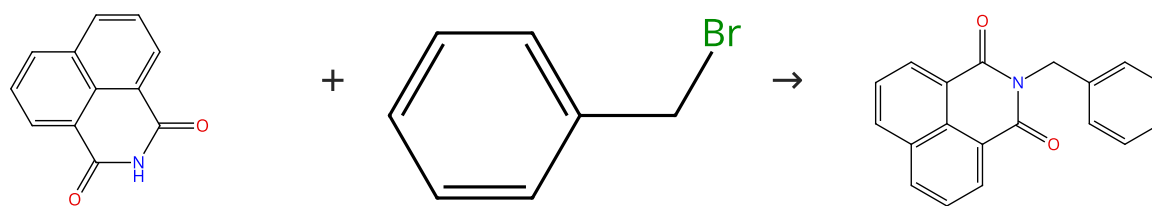
1.2 **Solvents:** Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (54)

Suppliers (67)

Suppliers (41)

31-041-CAS-18435823

Steps: 1 Yield: 98%

Mechanochemical N-alkylation of imides

1.1 Reagents: Potassium carbonate; 1 h

By: Bris, Anamarija; et al

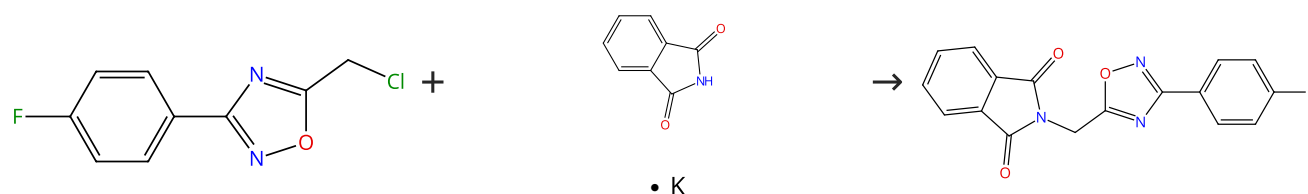
1.2 Solvents: Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (56)

Suppliers (93)

Suppliers (3)

31-041-CAS-8280559

Steps: 1 Yield: 97%

Design, Synthesis, and Biological Activities of Novel 2- Cyanoacrylates Containing Oxazole, Oxadiazole, or Quinoline Moieties

1.1 Solvents: Dimethylformamide; rt; 5 h, rt

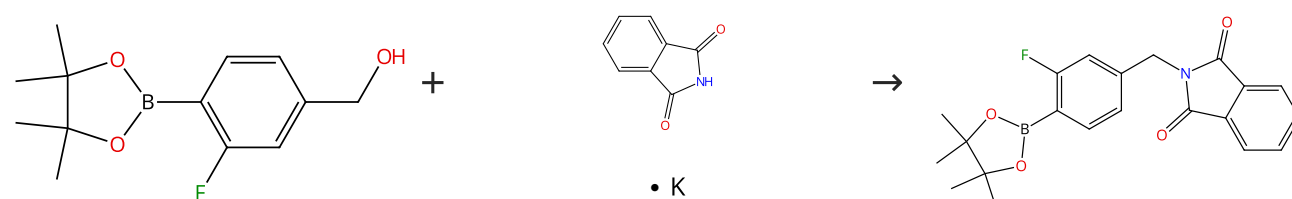
By: Zhao, Qiqi; et al

Experimental Protocols

Journal of Agricultural and Food Chemistry (2009), 57(7), 2849-2855.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (56)

Suppliers (93)

Suppliers (2)

31-041-CAS-4975820

Steps: 1 Yield: 97%

Synthesis of boron-containing primary amines

1.1 Reagents: Methanesulfonyl chloride, Diisopropylethylamine
Solvents: Dichloromethane; 0 °C; 3 h, 0 °C

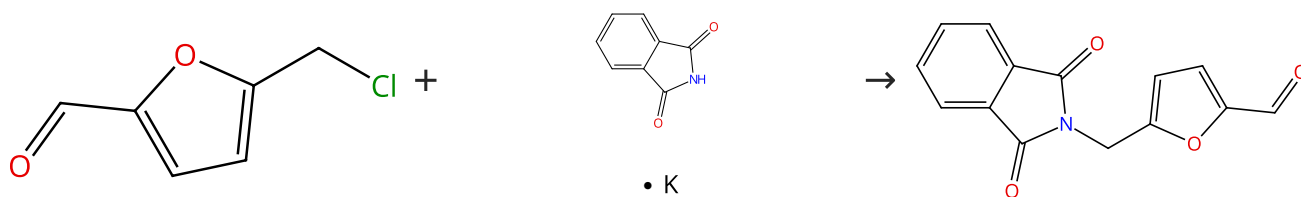
By: Chung, Sheng-Hsuan; et al

1.2 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 3 d, rt

Molecules (2013), 18(10), 12346-12367.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (61)

Suppliers (93)

Suppliers (37)

31-614-CAS-42744107

Steps: 1 Yield: 97%

A novel approach for the preparation of furandiamines utilizing biomass platform chemicals as substrates via Gabriel synthesis

1.1 **Solvents:** Ethanol, Dimethylformamide, Acetonitrile, Tetrahydrofuran, 1,4-Dioxane; 1 - 5 h, 30 °C → 70 °C

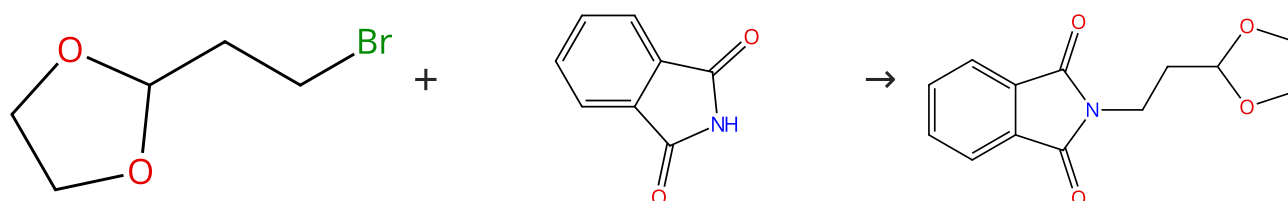
Experimental Protocols

By: Sun, Tong; et al

Molecular Catalysis (2025), 570, 114679.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (79)

Suppliers (82)

Suppliers (11)

31-041-CAS-2794673

Steps: 1 Yield: 96%

Synthesis of 2-(2-(2-phthalimidoethyl)-1,3-dioxolane

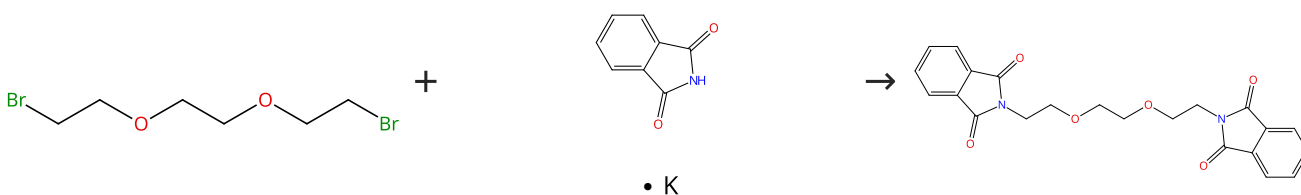
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrabutylammonium bromide, Potassium iodide
Solvents: Toluene; 3.5 h, 100 - 103 °C

By: Zeng, Xiang-chao; et al

Huaxue Shiji (2006), 28(5), 313-314.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (73)

Suppliers (93)

Suppliers (11)

31-041-CAS-23355615

Steps: 1 Yield: 96%

Study on synthesis of 1, 8- diamino-3, 6-dioxaoctane

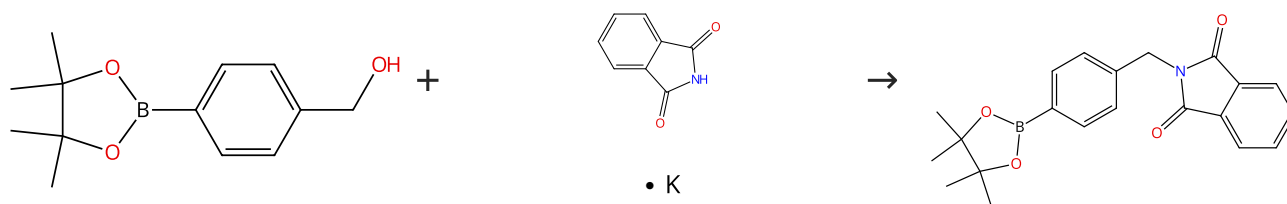
1.1 **Solvents:** Diethyl ether, Dimethylformamide; 100 °C → 120 °C; 8 h, 120 °C

By: Chen, Jianliang

Guangdong Huagong (2015), 42(17), 221-222.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (85)

Suppliers (93)

Suppliers (67)

31-041-CAS-7747028

Steps: 1 Yield: 96%

Synthesis of boron-containing primary amines

1.1 Reagents: Methanesulfonyl chloride, Diisopropylethylamine
Solvents: Dichloromethane; 0 °C; 3 h, 0 °C

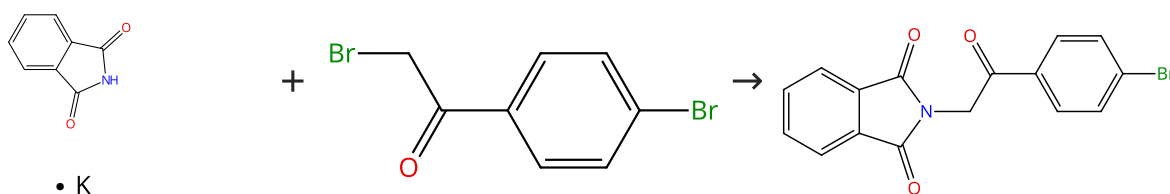
By: Chung, Sheng-Hsuan; et al

1.2 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 3 d, rt

Molecules (2013), 18(10), 12346-12367.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (93)

Suppliers (88)

Suppliers (19)

31-041-CAS-10226938

Steps: 1 Yield: 96%

Organic solid-state reactions. Solid-state Gabriel reactions

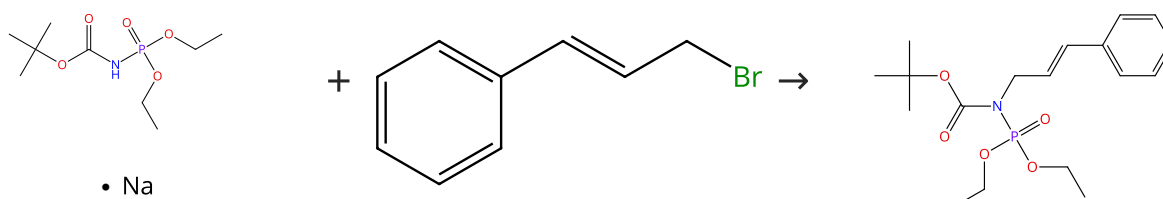
No Data Available

By: Tokuno, Kenji; et al

Yakugaku Zasshi (1991), 111(7), 359-64.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (2)

Suppliers (64)

31-041-CAS-15445375

Steps: 1 Yield: 96%

Diethyl N-(tert-butoxycarbonyl)phosphoramidate: a new, useful substitute of phthalimide in Gabriel-type syntheses of primary amines

No Data Available

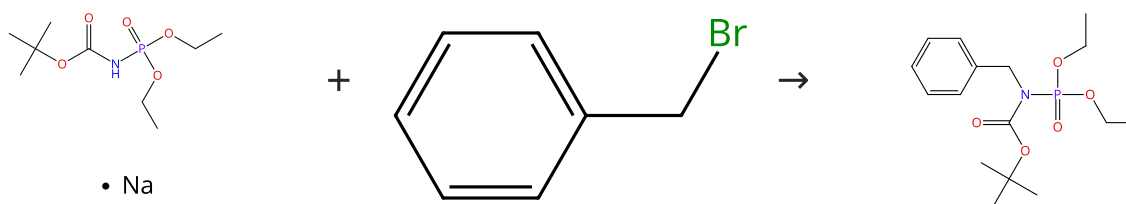
By: Zwierzak, Andrzej; et al

Experimental Protocols

Synthesis (1982), (11), 922-4.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (2)

Suppliers (67)

Suppliers (2)

31-041-CAS-15161141

Steps: 1 Yield: 95%

Diethyl N-(tert-butoxycarbonyl)phosphoramidate: a new, useful substitute of phthalimide in Gabriel-type syntheses of primary amines

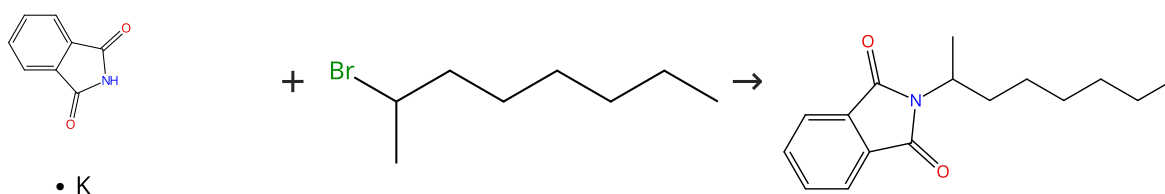
By: Zwierzak, Andrzej; et al
Synthesis (1982), (11), 922-4.

No Data Available

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (93)

Suppliers (59)

Suppliers (7)

31-041-CAS-16819421

Steps: 1 Yield: 95%

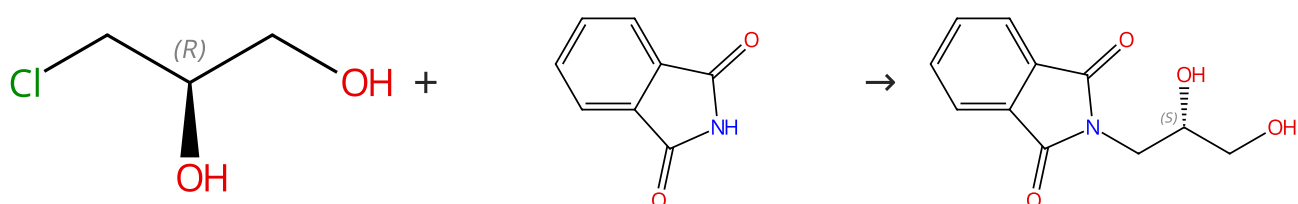
Phthalimide

1.1 **Catalysts:** Hexadecyltributylphosphonium bromide
Solvents: Toluene; 0.7 - 18 h, 25 - 100 °C

By: de Koning, Henk; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-3.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 95%



Absolute stereochemistry shown,
Rotation (-)

Suppliers (82)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (74)

Suppliers (17)

31-041-CAS-19431470

Steps: 1 Yield: 95%

Method for preparation of (αS)-1,3-dihydro-α-hydroxy-1,3-dioxo-2H-isoindole-2-propanoic acid

1.1 **Catalysts:** Potassium carbonate
Solvents: Dimethylformamide; rt → 110 °C; 5 h, 110 °C

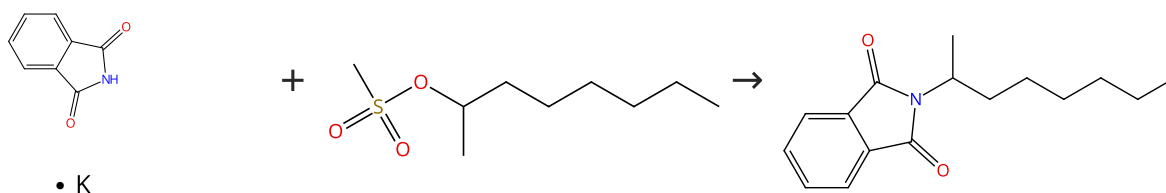
Assignees: Hisun Pharmaceutical Nantong Co., Ltd.; Zhejiang Hisun Pharmaceutical Co., Ltd.

China, CN108586313 A 2018-09-28

PatentPak available

Scheme 22 (1 Reaction)

Steps: 1 Yield: 95%



• K

Suppliers (93)

Suppliers (5)

Suppliers (7)

31-041-CAS-21063853

Steps: 1 Yield: 95%

Phthalimide

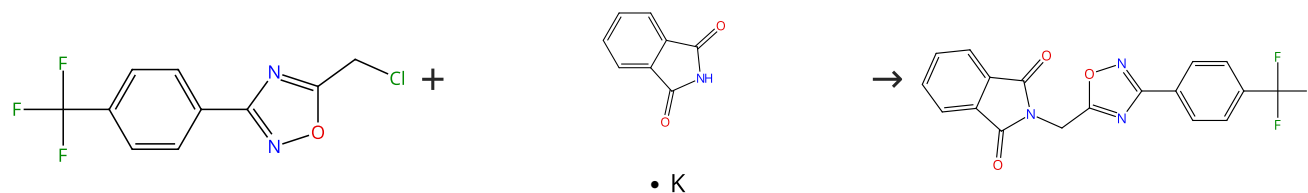
1.1 Catalysts: Hexadecyltributylphosphonium bromide
Solvents: Toluene; 0.7 - 18 h, 25 - 100 °C

By: de Koning, Henk; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis
(2001), 1-3.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 95%



• K

Suppliers (53)

Suppliers (93)

Supplier (1)

31-041-CAS-6145352

Steps: 1 Yield: 95%

Design, Synthesis, and Biological Activities of Novel 2- Cyanoacrylates Containing Oxazole, Oxadiazole, or Quinoline Moieties

1.1 Solvents: Dimethylformamide; rt; 5 h, rt

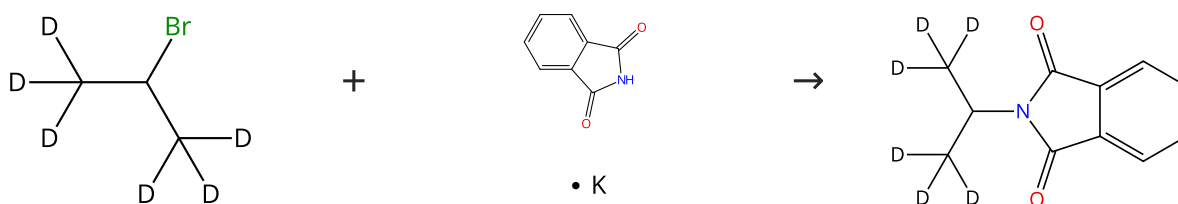
By: Zhao, Qiqi; et al

Experimental Protocols

Journal of Agricultural and Food Chemistry (2009), 57(7), 2849-2855.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 95%



• K

Suppliers (22)

Suppliers (93)

31-041-CAS-16229854

Steps: 1 Yield: 95%

Efficient synthesis of D₆-clenproperol and D₆-cimaterol using deuterium isopropylamine as labeled precursor

1.1 Solvents: Dimethylformamide; 4 h, 120 °C

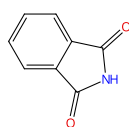
By: Sun, Kai; et al

Experimental Protocols

Journal of Labelled Compounds and Radiopharmaceuticals
(2016), 59(13), 552-556.

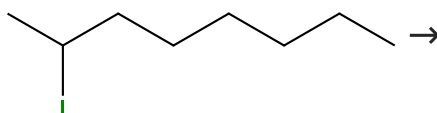
Scheme 25 (1 Reaction)

Steps: 1 Yield: 95%

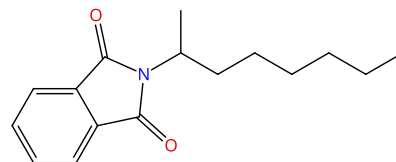


• K

+



→


 Suppliers (93)

 Suppliers (22)

 Suppliers (7)

31-041-CAS-16819423

Steps: 1 Yield: 95%

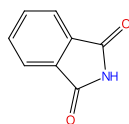
Phthalimide

1.1 Catalysts: Hexadecyltributylphosphonium bromide
Solvents: Toluene; 0.7 - 18 h, 25 - 100 °C

By: de Koning, Henk; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-3.

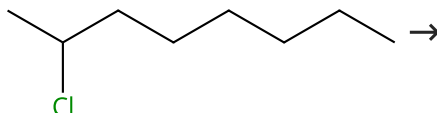
Scheme 26 (1 Reaction)

Steps: 1 Yield: 95%

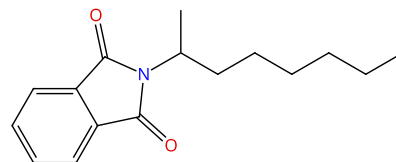


• K

+



→


 Suppliers (93)

 Suppliers (16)

 Suppliers (7)

31-041-CAS-16819415

Steps: 1 Yield: 95%

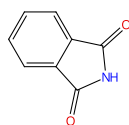
Phthalimide

1.1 Catalysts: Hexadecyltributylphosphonium bromide
Solvents: Toluene; 0.7 - 18 h, 25 - 100 °C

By: de Koning, Henk; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-3.

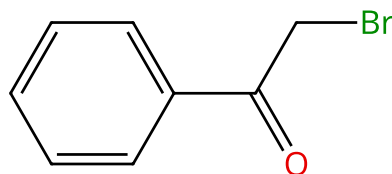
Scheme 27 (2 Reactions)

Steps: 1 Yield: 92-95%

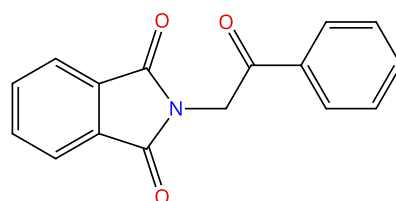


• K

+



→


 Suppliers (93)

 Suppliers (74)

 Suppliers (53)

31-041-CAS-3558046

Steps: 1 Yield: 95%

Microwave-assisted organic synthesis: the Gabriel approach as a route to new pyrazolylhydrazonoazoles

1.1 Solvents: Dimethylformamide; 30 min, heated

By: Al-Mousawi, Saleh M.; et al
ARKIVOC (Gainesville, FL, United States) (2008), (16), 268-278.

31-041-CAS-16819411

Steps: 1 Yield: 92%

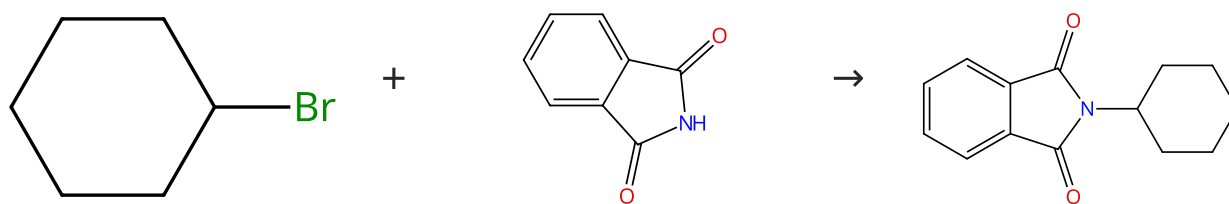
Phthalimide

1.1 Solvents: Dimethylformamide; 0.5 h, rt → 55 °C

By: de Koning, Henk; et al
e-EROS Encyclopedia of Reagents for Organic Synthesis (2001), 1-3.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (74)

Suppliers (82)

Suppliers (16)

31-041-CAS-7024237

Steps: 1 Yield: 95%

Effects of cryptands and activation of bases. V. Action of alkali hydrides on weak acids. II. Alkylation of anions obtained

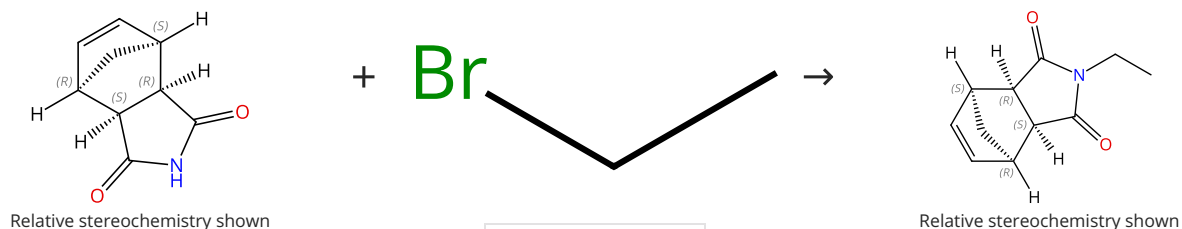
By: Pasquini, M. A.; et al

Tetrahedron (1980), 36(9), 1223-6.

No Data Available

Scheme 29 (1 Reaction)

Steps: 1 Yield: 95%



Relative stereochemistry shown

Relative stereochemistry shown

Suppliers (47)

Suppliers (54)

Suppliers (2)

31-041-CAS-18435820

Steps: 1 Yield: 95%

Mechanochemical N-alkylation of imides

By: Bris, Anamarija; et al

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

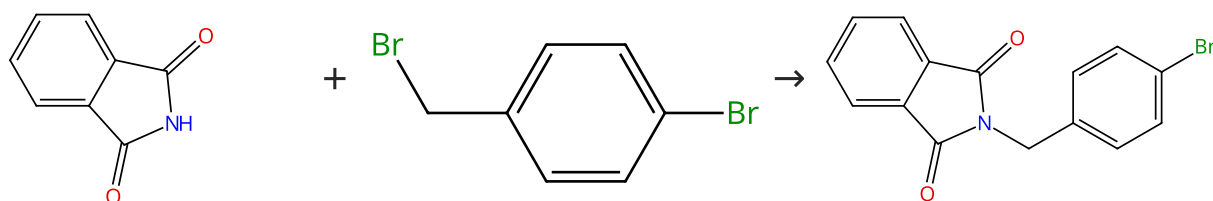
1.1 Reagents: Potassium carbonate; 1 h

1.2 Solvents: Dimethylformamide; 1 h

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (82)

Suppliers (76)

Suppliers (52)

31-041-CAS-18435827

Steps: 1 Yield: 95%

Mechanochemical N-alkylation of imides

By: Bris, Anamarija; et al

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

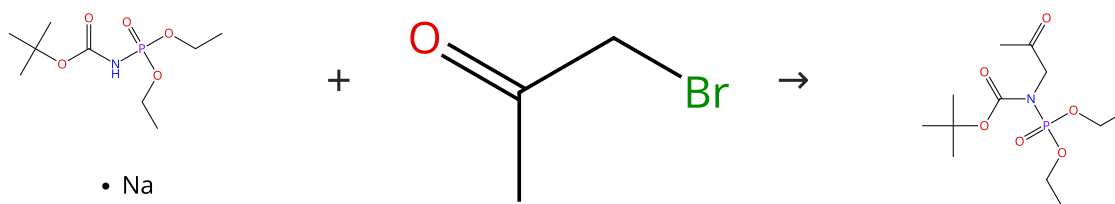
1.1 Reagents: Potassium carbonate; 1 h

1.2 Solvents: Dimethylformamide; 1 h

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (2)

Suppliers (28)

Supplier (1)

31-041-CAS-8448348

Steps: 1 Yield: 94%

Diethyl N-(tert-butoxycarbonyl)phosphoramidate: a new, useful substitute of phthalimide in Gabriel-type syntheses of primary amines

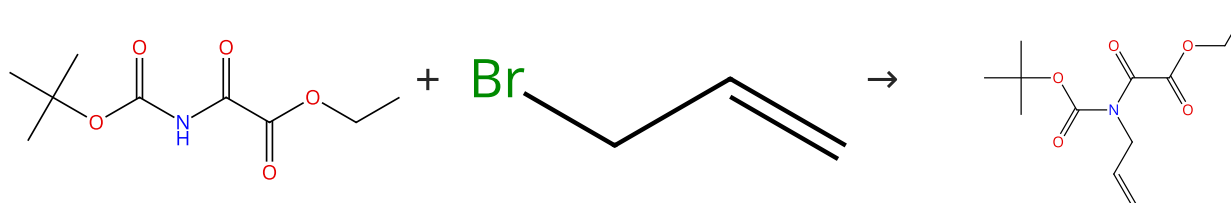
By: Zwierzak, Andrzej; et al
Synthesis (1982), (11), 922-4.

No Data Available

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (58)

Suppliers (57)

Supplier (1)

31-041-CAS-7522838

Steps: 1 Yield: 94%

N-Boc and N-Cbz ethyl oxamates: new Gabriel reagents

By: Berree, Fabienne; et al
Synthetic Communications (1999), 29(15), 2685-2693.

1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Dimethylformamide

1.2 **Solvents:** Dichloromethane

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (65)

Suppliers (86)

Suppliers (8)

31-041-CAS-10534487

Steps: 1 Yield: 94%

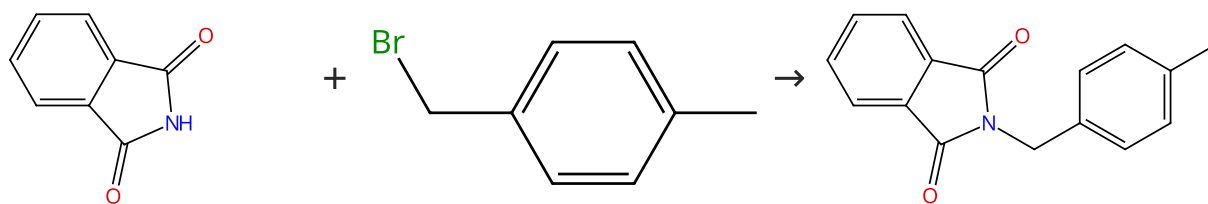
Triflamides for protection and monoalkylation of amines and a new Gabriel synthesis

By: Hendrickson, James B.; et al
Tetrahedron Letters (1973), (39), 3839-42.

No Data Available

Scheme 34 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (82)

Suppliers (56)

Suppliers (36)

31-041-CAS-18435826

Steps: 1 Yield: 94%

Mechanochemical N-alkylation of imides1.1 **Reagents:** Potassium carbonate; 1 h

By: Bris, Anamarija; et al

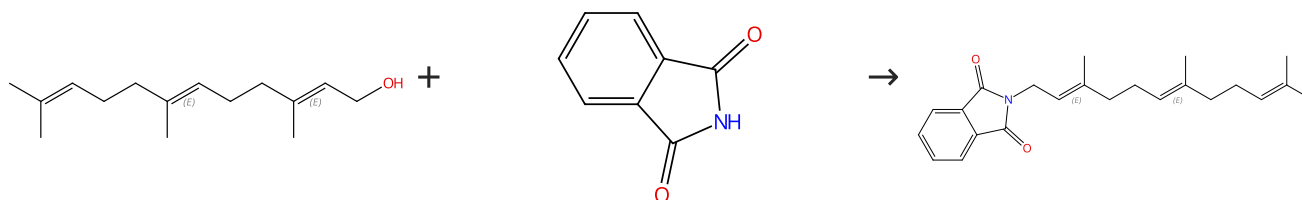
1.2 **Solvents:** Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 93%



Double bond geometry shown

Suppliers (71)

Suppliers (82)

Double bond geometry shown

Supplier (1)

31-041-CAS-8551009

Steps: 1 Yield: 93%

A convenient two-step procedure for the synthesis of substituted allylic amines from allylic alcohols1.1 **Reagents:** Triphenylphosphine, Diisopropyl azodicarboxylate
Solvents: Tetrahydrofuran

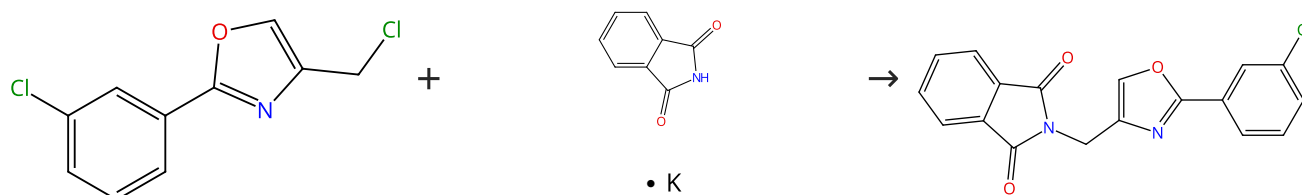
By: Sen, Stephanie E.; et al

Experimental Protocols

Synthesis (1995), (7), 756-8.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (30)

Suppliers (93)

Suppliers (2)

31-041-CAS-15271927

Steps: 1 Yield: 93%

Design, Synthesis, and Biological Activities of Novel 2-Cyanoacrylates Containing Oxazole, Oxadiazole, or Quinoline Moieties1.1 **Solvents:** Dimethylformamide; rt; 5 h, rt

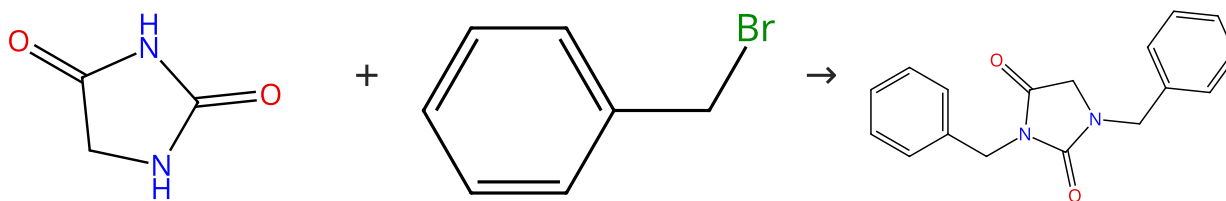
By: Zhao, Qiqi; et al

Experimental Protocols

Journal of Agricultural and Food Chemistry (2009), 57(7), 2849-2855.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (85)

Suppliers (67)

Suppliers (3)

31-041-CAS-18435833

Steps: 1 Yield: 93%

Mechanochemical N-alkylation of imides

1.1 Reagents: Potassium carbonate; 1 h

By: Bris, Anamarija; et al

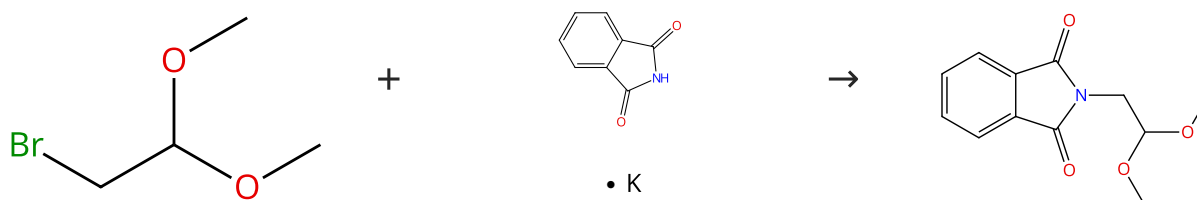
1.2 Solvents: Dimethylformamide; 1 h

Beilstein Journal of Organic Chemistry (2017), 13, 1745-1752.

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (81)

Suppliers (93)

Suppliers (25)

31-041-CAS-9953991

Steps: 1 Yield: 93%

Synthesis of 2,2-dimethoxyethanamine

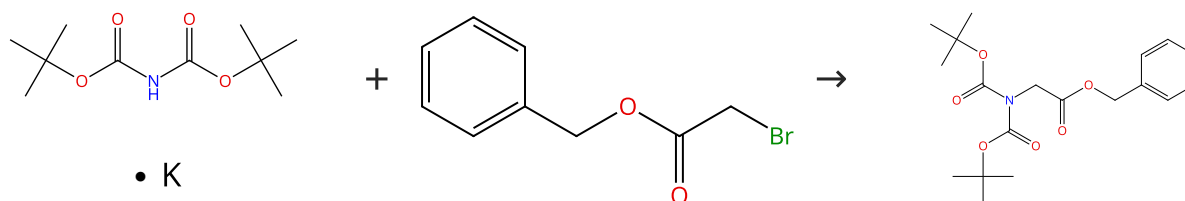
1.1 Solvents: Dimethylformamide; rt → 140 °C; 4 h, 140 °C

By: Jin, Jiyu; et al

Jingxi Huagong Zhongjianti (2011), 41(2), 36-37, 55.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (42)

Suppliers (86)

Suppliers (3)

31-041-CAS-4304887

Steps: 1 Yield: 93%

A convenient one-flask preparation of di-tert-butyl iminodibarbonate: a versatile Gabriel reagent

1.1 Solvents: Dimethylformamide

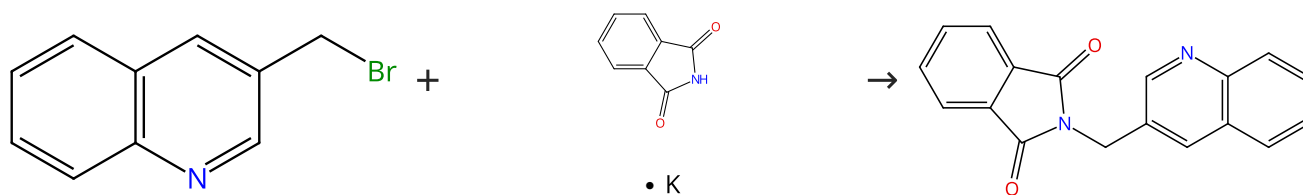
By: Grehn, Leif; et al

Experimental Protocols

Synthesis (1987), (3), 275-6.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (36)

Suppliers (93)

Suppliers (6)

31-041-CAS-4393787

Steps: 1 Yield: 92%

Design, Synthesis, and Biological Activities of Novel 2- Cyanoacrylates Containing Oxazole, Oxadiazole, or Quinoline Moieties

By: Zhao, Qiqi; et al

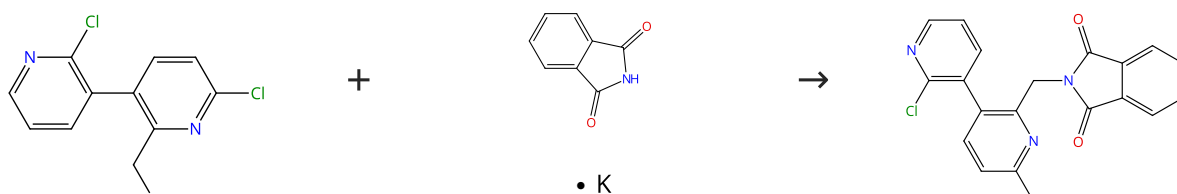
Journal of Agricultural and Food Chemistry (2009), 57(7), 2849-2855.

1.1 Solvents: Dimethylformamide; rt; 5 h, rt

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (93)

31-041-CAS-15888962

Steps: 1 Yield: 92%

Synthesis and Reactivity of Triazaphenanthrenes

By: Fernandez, Sarah; et al

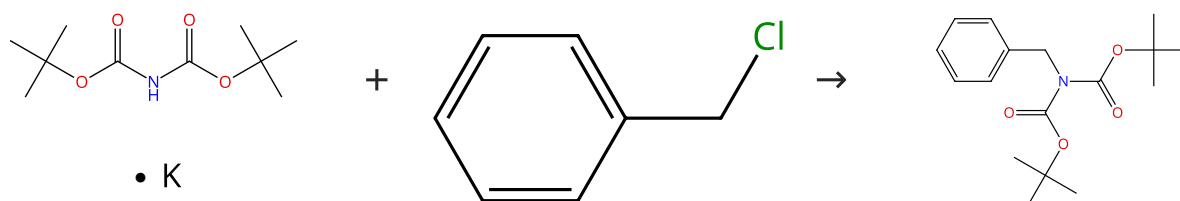
Organic Letters (2016), 18(13), 3158-3161.

1.1 Solvents: Dimethylformamide; 25 °C; 25 °C → 100 °C; 4 h, 100 °C

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (42)

Suppliers (65)

Suppliers (29)

31-041-CAS-2208753

Steps: 1 Yield: 92%

A convenient one-flask preparation of di-tert-butyl iminodibutylate: a versatile Gabriel reagent

By: Grehn, Leif; et al

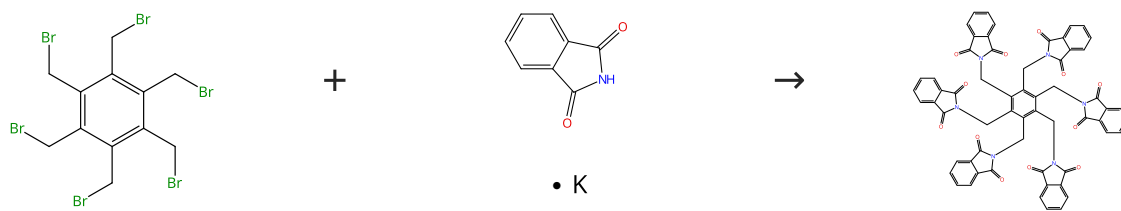
Synthesis (1987), (3), 275-6.

1.1 Solvents: Dimethylformamide

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (53)

Suppliers (93)

31-041-CAS-19437697

Steps: 1 Yield: 92%

Gabriel Synthesis of Hexakis(aminomethyl)benzene and Its Derivatization

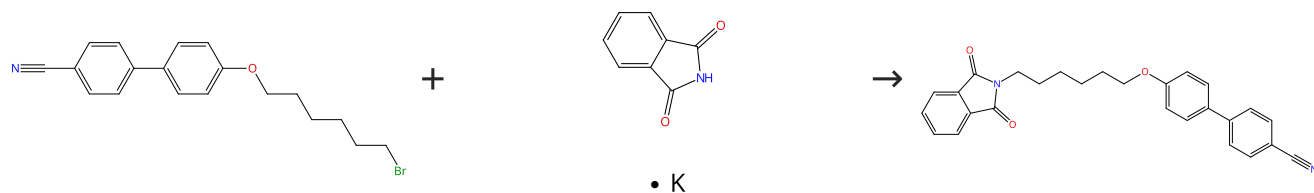
1.1 Solvents: Dimethylformamide; 18 h, 70 °C

By: Masuda, Junya; et al

ChemistrySelect (2018), 3(22), 6112-6115.

Scheme 44 (1 Reaction)

Steps: 1 Yield: 92%



Supplier (1)

Suppliers (93)

31-041-CAS-12824480

Steps: 1 Yield: 92%

Synthesis of 4-(6-aminohexyloxy)-4'-cyanobiphenyl

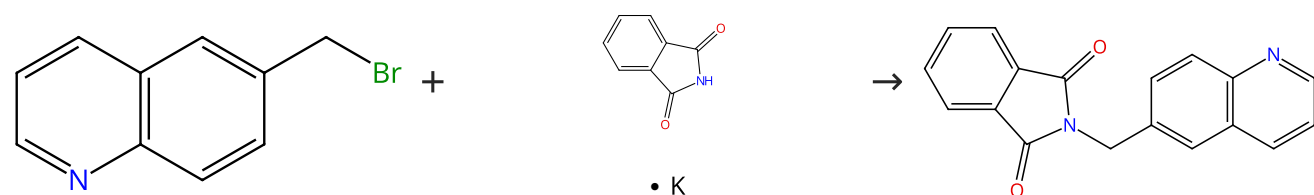
1.1 Solvents: Dimethylformamide; 80 °C; 6 h, 80 °C

By: Zhang, Cai-yun; et al

Jingxi Huagong (2005), 22(11), 806-808.

Scheme 45 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (34)

Suppliers (93)

Suppliers (5)

31-041-CAS-6428732

Steps: 1 Yield: 92%

Design, Synthesis, and Biological Activities of Novel 2- Cyanoacrylates Containing Oxazole, Oxadiazole, or Quinoline Moieties

1.1 Solvents: Dimethylformamide; rt; 5 h, rt

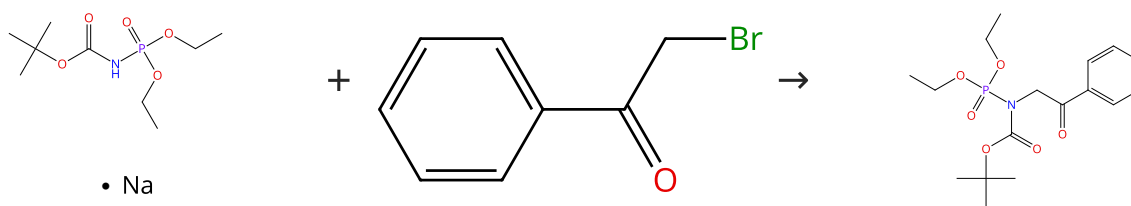
By: Zhao, Qiqi; et al

Experimental Protocols

Journal of Agricultural and Food Chemistry (2009), 57(7), 2849-2855.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (2)

Suppliers (74)

Supplier (1)

31-041-CAS-12695515

Steps: 1 Yield: 92%

No Data Available

Experimental Protocols

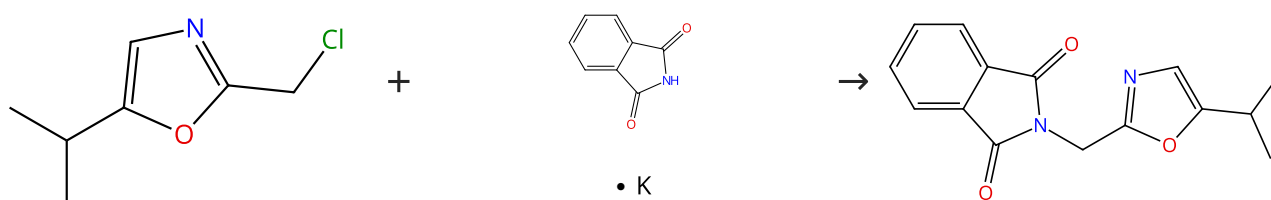
Diethyl N-(tert-butoxycarbonyl)phosphoramidate: a new, useful substitute of phthalimide in Gabriel-type syntheses of primary amines

By: Zwierzak, Andrzej; et al

Synthesis (1982), (11), 922-4.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (24)

Suppliers (93)

Suppliers (3)

31-041-CAS-1889866

Steps: 1 Yield: 91%

1.1 Solvents: Dimethylformamide; rt; 5 h, rt

Experimental Protocols

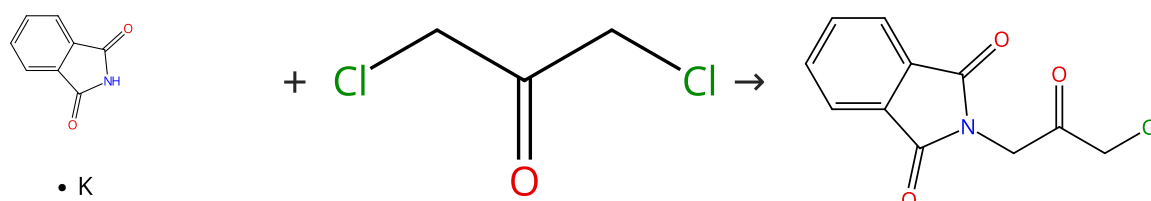
Design, Synthesis, and Biological Activities of Novel 2- Cyanoacrylates Containing Oxazole, Oxadiazole, or Quinoline Moieties

By: Zhao, Qiqi; et al

Journal of Agricultural and Food Chemistry (2009), 57(7), 2849-2855.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (93)

Suppliers (56)

Suppliers (47)

31-041-CAS-7648341

Steps: 1 Yield: 91%

1.1 Solvents: Water; rt; 1 h, rt

Process for synthesis of 5- aminolevulinate hydrochloride

Assignee: Shandong University

China, CN102627573 A 2012-08-08

PatentPak available